

UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education
Advanced Subsidiary Level and Advanced Level

MATHEMATICS

9709/04

Paper 4 Mechanics 1 **(M1)**

October/November 2005

1 hour 15 minutes

Additional materials: Answer Booklet/Paper
Graph paper
List of Formulae (MF9)

READ THESE INSTRUCTIONS FIRST

If you have been given an Answer Booklet, follow the instructions on the front cover of the Booklet.
Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.
Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.
Where a numerical value for the acceleration due to gravity is needed, use 10 m s^{-2} .
At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.
The total number of marks for this paper is 50.
Questions carrying smaller numbers of marks are printed earlier in the paper, and questions carrying larger numbers of marks later in the paper.
The use of an electronic calculator is expected, where appropriate.
You are reminded of the need for clear presentation in your answers.

This document consists of 4 printed pages.



- 1 A car travels in a straight line with constant acceleration $a \text{ m s}^{-2}$. It passes the points A , B and C , in this order, with speeds 5 m s^{-1} , 7 m s^{-1} and 8 m s^{-1} respectively. The distances AB and BC are $d_1 \text{ m}$ and $d_2 \text{ m}$ respectively.

(i) Write down an equation connecting

(a) d_1 and a ,

(b) d_2 and a .

[2]

(ii) Hence find d_1 in terms of d_2 .

[2]

- 2 A crate of mass 50 kg is dragged along a horizontal floor by a constant force of magnitude 400 N acting at an angle α° upwards from the horizontal. The total resistance to motion of the crate has constant magnitude 250 N . The crate starts from rest at the point O and passes the point P with a speed of 2 m s^{-1} . The distance OP is 20 m . For the crate's motion from O to P , find

(i) the increase in kinetic energy of the crate,

[1]

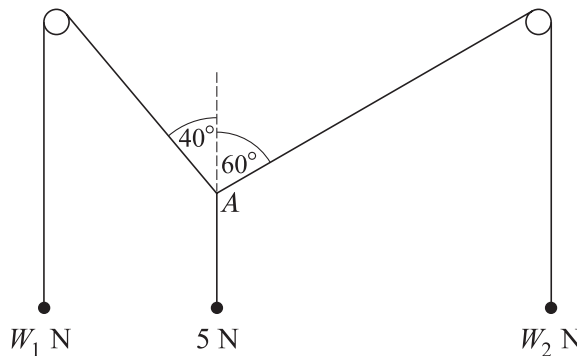
(ii) the work done against the resistance to the motion of the crate,

[1]

(iii) the value of α .

[3]

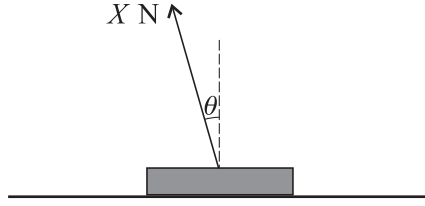
3



Each of three light strings has a particle attached to one of its ends. The other ends of the strings are tied together at a point A . The strings are in equilibrium with two of them passing over fixed smooth horizontal pegs, and with the particles hanging freely. The weights of the particles, and the angles between the sloping parts of the strings and the vertical, are as shown in the diagram. Find the values of W_1 and W_2 .

[6]

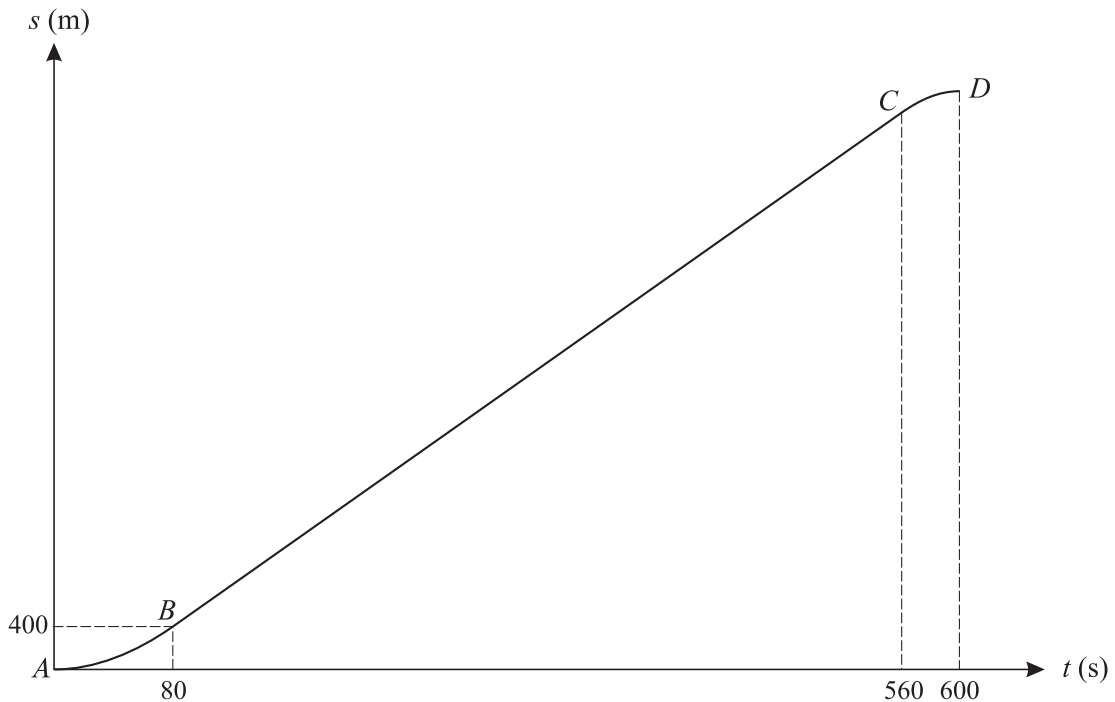
4



A stone slab of mass 320 kg rests in equilibrium on rough horizontal ground. A force of magnitude $X \text{ N}$ acts upwards on the slab at an angle of θ to the vertical, where $\tan \theta = \frac{7}{24}$ (see diagram).

- (i) Find, in terms of X , the normal component of the force exerted on the slab by the ground. [3]
- (ii) Given that the coefficient of friction between the slab and the ground is $\frac{3}{8}$, find the value of X for which the slab is about to slip. [3]

5



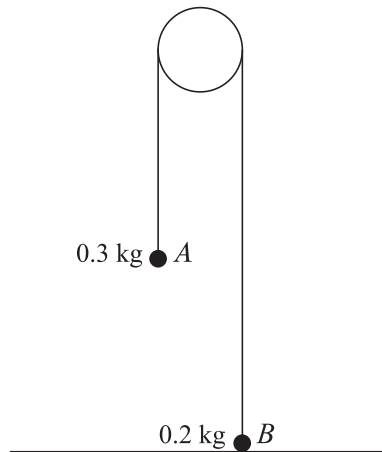
The diagram shows the displacement-time graph for a car's journey. The graph consists of two curved parts AB and CD , and a straight line BC . The line BC is a tangent to the curve AB at B and a tangent to the curve CD at C . The gradient of the curves at $t = 0$ and $t = 600$ is zero, and the acceleration of the car is constant for $0 < t < 80$ and for $560 < t < 600$. The displacement of the car is 400 m when $t = 80$.

- (i) Sketch the velocity-time graph for the journey. [3]
- (ii) Find the velocity at $t = 80$. [2]
- (iii) Find the total distance for the journey. [2]
- (iv) Find the acceleration of the car for $0 < t < 80$. [2]

- 6 A particle P starts from rest at O and travels in a straight line. Its velocity $v \text{ m s}^{-1}$ at time $t \text{ s}$ is given by $v = 8t - 2t^2$ for $0 \leq t \leq 3$, and $v = \frac{54}{t^2}$ for $t > 3$. Find

- (i) the distance travelled by P in the first 3 seconds, [4]
- (ii) an expression in terms of t for the displacement of P from O , valid for $t > 3$, [3]
- (iii) the value of v when the displacement of P from O is 27 m. [3]

7



Two particles A and B , of masses 0.3 kg and 0.2 kg respectively, are attached to the ends of a light inextensible string which passes over a smooth fixed pulley. Particle B is held on the horizontal floor and particle A hangs in equilibrium. Particle B is released and each particle starts to move vertically with constant acceleration of magnitude $a \text{ m s}^{-2}$.

- (i) Find the value of a . [4]

Particle A hits the floor 1.2 s after it starts to move, and does not rebound upwards.

- (ii) Show that A hits the floor with a speed of 2.4 m s^{-1} . [1]
- (iii) Find the gain in gravitational potential energy by B , from leaving the floor until reaching its greatest height. [5]